

ORAVITA, Romania

WMO: 153380

Lat: 45.0386N

Lon: 21.7106E

Elev: 310

StdP: 97.65

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.0	-13.1	1.2	-6.1	-10.8	1.5	-4.4	20.4	-0.4	16.0	0.8	1.6	320	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	32.8	20.3	30.9	20.5	29.2	20.2	22.6	29.1	21.7	28.2	20.8	27.4	2.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.6	26.4	19.4	14.7	25.3	18.6	14.0	24.2	68.0	29.1	64.6	28.0	61.6	27.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	10.3	8.2	-13.0	34.5	3.0	2.5	-15.2	36.3	-16.9	37.8	-18.6	39.2	-20.8	41.0		
(5)			-13.4	24.2	2.7	1.6	-15.4	25.3	-17.0	26.3	-18.5	27.2	-20.5	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	1.1	3.8	6.8	12.2	16.7	20.4	22.2	22.5	17.7	12.9	7.8	2.4		
	DBStd	8.79	5.19	5.12	4.71	5.09	4.04	3.65	3.51	4.05	4.66	5.06	5.51	4.90		
	HDD10.0	997	278	182	120	36	2	0	0	0	1	31	106	241		
	HDD18.3	2665	534	408	358	191	79	21	7	9	67	179	317	494		
	CDD10.0	1820	3	7	21	103	211	311	379	389	232	119	41	4		
	CDD18.3	446	0	0	1	7	29	82	128	140	48	10	1	0		
	CDH23.3	3483	0	0	1	28	147	613	1091	1222	348	32	2	0		
	CDH26.7	1121	0	0	0	1	13	160	381	463	101	2	0	0		
(14) Wind	WSAvg	2.6	2.4	2.7	2.9	3.3	2.7	2.0	2.0	2.4	2.8	2.7	3.0	2.3		
(15) Precipitation	PrecAvg	767	49	43	44	66	78	92	96	58	72	60	52	56		
	PrecMax	995	97	110	119	120	156	207	222	121	189	131	140	145		
	PrecMin	443	12	2	12	22	23	11	7	7	12	2	0	10		
	PrecStd	153	24	27	27	25	34	51	58	34	40	34	30	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	17.6	21.3	26.1	28.2	31.8	34.7	35.3	32.9	26.3	21.4	15.9		
		MCWB	9.0	10.3	12.6	15.5	19.0	21.4	20.3	20.9	19.2	16.2	13.1	10.1		
	2%	DB	12.7	15.0	18.6	23.7	26.6	30.1	32.5	33.2	29.2	23.9	18.8	13.6		
		MCWB	7.7	9.2	10.7	14.3	18.1	21.1	20.5	20.2	19.1	15.8	12.5	8.7		
	5%	DB	10.6	13.3	16.5	21.7	25.0	28.5	30.4	31.2	27.0	21.9	17.5	11.6		
		MCWB	6.5	8.3	9.7	13.3	17.2	20.5	20.6	20.4	18.0	14.8	12.1	7.4		
	10%	DB	8.7	11.3	14.2	19.7	23.4	27.0	28.6	29.1	25.0	20.0	15.7	9.2		
		MCWB	5.2	6.9	8.5	12.4	16.2	19.8	20.3	20.0	17.3	14.4	11.0	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	11.1	13.0	16.7	20.2	23.7	23.6	23.7	21.3	18.9	13.6	11.0		
		MCDB	14.5	16.2	21.3	24.0	27.2	29.5	30.3	31.3	29.7	23.9	19.3	13.9		
	2%	WB	8.2	9.6	11.2	15.2	19.0	22.3	22.4	22.3	19.8	16.8	13.0	9.1		
		MCDB	12.1	14.2	17.6	22.1	25.5	27.8	28.9	29.5	28.2	21.3	18.3	13.4		
	5%	WB	6.9	8.5	10.1	13.9	17.8	21.1	21.5	21.4	18.8	15.8	12.4	7.5		
		MCDB	10.1	12.7	15.8	20.4	23.9	27.0	28.0	28.5	25.3	20.7	17.0	10.9		
	10%	WB	5.6	7.3	8.8	12.9	16.7	20.3	20.7	20.5	17.7	14.7	11.4	6.3		
		MCDB	8.2	11.1	13.6	18.8	22.2	26.0	27.1	27.4	23.4	19.4	15.4	9.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.9	8.0	8.4	8.7	9.2	9.8	9.5	8.3	7.5	6.0	5.3		
		MCDBR	8.2	9.3	10.7	10.1	9.9	10.7	11.3	11.2	10.2	9.5	7.8	7.9		
	5% WB	MCWBR	4.9	5.3	5.9	5.1	5.2	5.3	4.9	4.4	4.3	4.7	3.9	4.7		
		MCWBR	7.5	8.5	9.7	10.0	9.5	10.5	10.3	10.5	9.2	8.7	7.3	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.361	0.399	0.455	0.441	0.433	0.442	0.457	0.427	0.387	0.355	0.331		
		taud	2.313	2.225	2.197	2.061	2.161	2.223	2.199	2.155	2.206	2.314	2.355	2.347		
	Ebn at Noon		761	804	824	806	832	839	825	795	785	767	730	722		
		Edh at Noon	81	106	125	156	146	138	140	141	123	97	78	72		
(44) All-Sky Solar Radiation	RadAvg		1.23	2.06	3.29	4.55	5.50	6.12	6.14	5.48	3.84	2.61	1.56	1.03		
	RadStd		0.15	0.28	0.43	0.47	0.52	0.57	0.47	0.53	0.41	0.30	0.16	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		+0.65	N/A	N/A	N/A	+0.36	N/A	N/A	-157	+109	+60

Nomenclature: See separate page